Grasping-region expansion using rear-pole magnets in electromagnetic grippers

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***Abstract*—This paper presents a geometric and magnetic analysis of grasping-region expansion in electromagnetic grippers using rear-pole permanent magnets. The gripper uses liquid-cooled laminar electromagnets (LCLEs) as active fingertips and permanent magnets placed behind them to modify the initial grasping posture. When the LCLEs are energized, the magnetic attraction between each LCLE and its rear-pole magnet pushes the fingertip forward and bends the carbon-fiber flexures connected to the finger structure. This coupled magnetic–flexural effect increases the maximum magnetic capture distance between the LCLE fingertips from approximately 23 mm to 50 mm, reducing insertion interference before the final grasp is established. The grasped handle does not need to be ferromagnetic; instead, it is constrained by the fingertip geometry while electromagnetic attraction shapes the finger posture and holding configuration. A simplified model is developed to relate rear-magnet attraction, carbon-fiber deflection, finger opening, handle geometry, load eccentricity, effective LCLE–LCLE interaction area, magnetic interaction, and grasping force. Experimental observations with circular, rectangular, and triangular handles are used to discuss how handle shape affects contact posture, pulling force, and sensor-based weight measurement. The proposed analysis provides design insight for electromagnetic grippers intended to grasp objects with handles of different materials, sizes, and geometries.**

***Index Terms*—Electromagnetic gripper, soft gripper, magnetic grasping and handle geometry.**

I. Introduction

Robotic grasping of heavy objects remains challenging as the gripper must generate sufficient holding force while remaining compact, lightweight, and adaptable to object geometry [1], [2]. Many everyday and industrial objects are manipulated through handles, whose diameter, cross-sectional shape, and contact condition can vary significantly [2]. Rigid grippers can provide repeatable contact kinematics, but their ability to adapt to different handle geometries is limited. Soft and compliant grippers can improve geometric adaptation, although their flexibility may reduce load capacity and make the contact condition more difficult to predict [2], [4].

Electromagnetic actuation is attractive for compact gripper mechanisms as it can generate large internal forces without requiring tendons, gears, screws, or other mechanical transmissions at the fingertip interface [3], [5]–[7]. Previous works have also explored the use of electromagnets and permanent magnets to improve grasping, locking, or motion control in robotic fingers and soft grippers [3]–[8]. However, in the gripper studied here, the grasped handle does not need to be ferromagnetic. Instead, the handle is mechanically constrained by the fingertip geometry, while electromagnetic attraction shapes the finger posture, closure behavior, and internal magnetic interaction of the gripper.

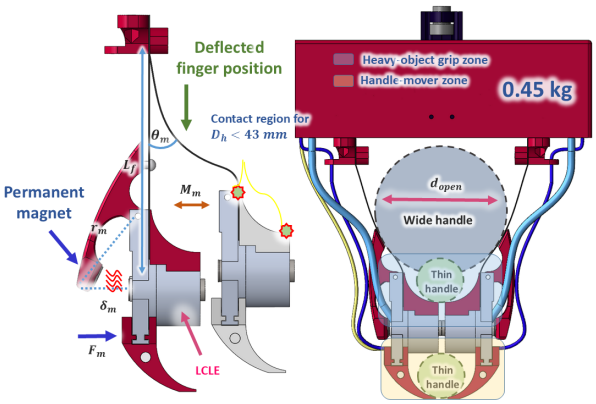


Fig. 1. Concept of grasping-region expansion using rear-pole permanent magnets. The attraction between the LCLE and the permanent magnet pushes the fingertip forward and bends the carbon-fiber flexure, increasing the initial opening available for mechanically grasping handles of different geometries.

Our previous studies introduced the LCLE actuator and characterized its electromagnetic and thermal performance [1], followed by the development of a flexible electromagnetic gripper for heavy-object manipulation [2]. Unlike these works, the present study specifically investigates how rear-pole permanent magnets expand the grasping region through carbon-fiber deformation, achieving a 117% increase in maximum magnetic capture distance. It also analyzes how handle geometry influences finger posture, pulling force, and weight-measurement accuracy.

A key feature of the gripper mechanism is the use of permanent magnets placed at the rear pole of each LCLE fingertip, as shown in Fig. 1. When the LCLE is energized, the magnetic attraction between the LCLE and the rear-pole magnet generates a mechanical action on the finger structure. Since the magnet holder is mechanically connected to the carbon-fiber flexures, this attraction pushes the LCLE forward and bends the carbon fibers. As a result, the initial separation between the fingertips increases, creating a larger available

region for receiving handles before the final grasping contact is established. In the studied mechanism, this effect increases the maximum magnetic capture distance between the LCLE fingertips from approximately 23 mm, when only the LCLE–LCLE attraction is used, to approximately 50 mm when the rear-pole magnets are included. This larger capture distance reduces the need to force the gripper against the handle during insertion.

This coupled magnetic–flexural effect allows larger handles to be inserted by increasing the initial finger separation. However, larger or non-circular handles can also modify fingertip alignment, LCLE–LCLE interaction area, effective core gap, and load-transfer position. Thus, grasping behavior depends on the combined effects of the rear-magnet mechanism, carbon-fiber bending, handle geometry, and load eccentricity.

This paper presents a geometric and magnetic analysis of grasping-region expansion using rear-pole magnets in electromagnetic grippers. The main objective is to explain how the rear-pole magnets induce carbon-fiber bending, increase the initial grasping region, reduce insertion interference, and modify the interaction between the gripper and handles of different geometries. Circular, rectangular, and triangular handles are considered to analyze how handle size and shape affect contact posture, pulling force, and sensor-based weight measurement.

II. LCLE-BASED ELECTROMAGNETIC GRIPPER WITH REAR-POLE MAGNETS

A. LCLE-based gripper architecture

The gripper uses two LCLE-based electromagnetic fingertips that generate attraction for finger closure, contact stabilization, and rear-magnet-induced deflection. The handle is grasped through mechanical constraint between the fingertip regions, so it does not need to be ferromagnetic. Since the LCLE actuator was previously developed for high holding force with internal oil cooling, this paper treats it as an existing platform and focuses on the interaction between the fingertip mechanism and object geometry.

The two fingertips are connected to the gripper body through flexible carbon-fiber elements. These elements provide structural support while allowing controlled bending during fingertip motion. This flexibility is essential for adapting the gripper to handles with different diameters and shapes. Fig. 2 summarizes the functional sequence of the gripper. The mechanism includes two moving elements that define the maximum handle diameter in the initial position. When the LCLEs are energized, the magnet–LCLE attraction bends the carbon-fiber supports and moves the fingertips toward a deflected position. Then, the LCLE–LCLE attraction closes the fingertips around the handle, producing distinct grasping regions for heavy-object and light-load grasping.

B. Rear-pole magnet mechanism

An N48 neodymium permanent magnet (3 mm diameter and 2 mm thickness) is placed behind each LCLE fingertip, as shown in Figs. 1 and 2. The LCLEs have a maximum operating current of 21 A each, while their electromagnetic

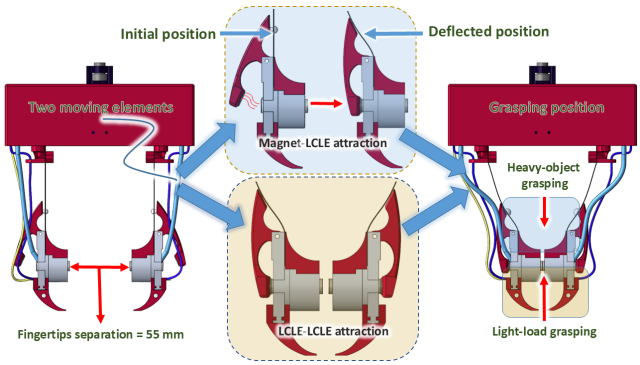


Fig. 2. Functional sequence of the electromagnetic fingertip gripper. In the initial position, the two moving elements define the maximum handle diameter that can be inserted. When the LCLEs are energized, the magnet–LCLE attraction bends the carbon-fiber supports and moves the fingertips toward a deflected position. The subsequent LCLE–LCLE attraction closes the fingertips around the handle, producing different grasping regions for heavy-object and light-load grasping.

and thermal characteristics were previously reported in [1], [2]. When the LCLE is energized, the magnetic interaction between the LCLE and the rear-pole magnet produces an attraction force F_m. Since the rear magnet is mounted on a mechanical holder connected to the carbon-fiber flexure, this force acts through an effective lever arm r_m and produces a bending action on the finger structure.

This interaction has two coupled effects. First, it pushes the LCLE fingertip forward, increasing the initial separation between the fingers. Second, it bends the carbon-fiber support, producing a magnet-induced deflection that enlarges the region available for handle insertion before the final grasping contact is established. Therefore, the rear-pole magnet is not only used as a magnetic element, but also as a mechanism for modifying the initial grasping geometry of the gripper. The maximum magnetic capture distance was measured using a ruler by progressively reducing the initial separation between the LCLE fingertips, energized at maximum power, until magnetic attraction initiated closure. The same procedure was applied with and without rear-pole permanent magnets, resulting in maximum capture distances of approximately 50 mm and 23 mm, respectively. This represents an increase of about 117%.

III. GEOMETRIC AND FLEXURAL MODEL OF GRASPING-REGION EXPANSION

A. Carbon-fiber flexure model

The carbon-fiber elements are modeled as flexible beams that support the LCLE fingertip and allow the grasping region to expand when the rear-pole permanent magnet is attracted by the energized LCLE. Each carbon-fiber flexure is 200 mm long, 18 mm wide, and 0.5 mm thick, with effective length L_f, width b_f, thickness h_f, and elastic modulus E_f. Since the elastic modulus of the employed temperature-resistant carbon-fiber composite was not experimentally characterized, E_f is retained as a model parameter. The second moment of area of the rectangular cross section is

$$I_f = \frac{b_f h_f^3}{12}. \tag{1}$$

The total flexural response of the finger is produced by two main contributions: the magnet-induced deflection generated by the rear-pole magnet mechanism and the object-load deflection generated during manipulation. Therefore, the finger deflection can be expressed as

$$\delta_f(D_h) = \delta_m + \delta_o(D_h), \tag{2}$$

where δ_m is the deflection induced by the magnet–LCLE interaction before or during closure, and $\delta_o(D_h)$ is the additional deflection caused by the load transmitted by a handle of diameter D_h.

When the LCLE is energized, the attraction force F_m between the LCLE and the rear-pole permanent magnet acts with an effective lever arm r_m. The resulting magnet-induced moment on the finger structure is approximated as

$$M_m = r_m F_m. \tag{3}$$

Assuming small deflections and modeling the carbon-fiber support as a cantilever beam subjected to an equivalent end moment, the magnet-induced tip deflection can be written as

$$\delta_m = \frac{M_m L_f^2}{2E_f I_f}. \tag{4}$$

The corresponding angular deflection is

$$\theta_m = \frac{M_m L_f}{E_f I_f}. \tag{5}$$

The magnet-induced deflection increases the initial opening available for handle insertion. The cantilever model assumes linear elasticity and small deflections; therefore, it provides a simplified description of the flexural behavior rather than an exact prediction, particularly at larger rotations. The experimentally observed capture-distance expansion demonstrates the mechanism's functionality but does not constitute a quantitative validation of the beam model. After the handle is grasped, the object load produces an additional bending contribution. If the vertical load transmitted to one finger is approximated as $F_o/2$, where

$$F_o = W_{\text{adjust}} g, \tag{6}$$

where W_{adjust} represents the object mass obtained from the load-cell measurement and used for subsequent weight estimation, and g is the gravitational acceleration. The resulting vertical-load deflection component can be estimated as

the vertical-load deflection component can be estimated as

$$\delta_{o,v} = \frac{(F_o/2)L_f^3}{3E_f I_f}. \tag{7}$$

This vertical-load term represents the bending caused by the object weight after the handle is supported by the gripper.

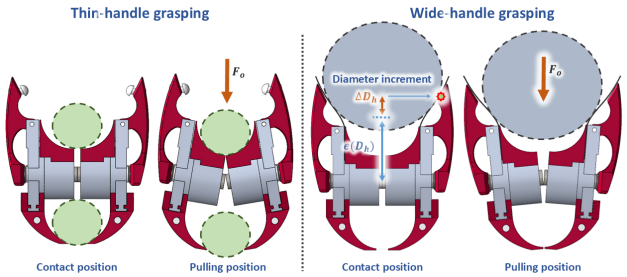


Fig. 3. Effect of handle diameter on the opening side of the LCLE–LCLE interface during manipulation. For thin handles, the load acts below the neutral interaction line, tending to open the interface from the lower side during pulling. For wider handles, the load acts above the neutral interaction line, tending to open the interface from the upper side.

However, the handle diameter also changes the load-transfer geometry, which introduces an additional moment during manipulation, as described in the next subsection.

B. Rear-magnet-induced finger opening

Let d_0 be the nominal initial separation between the fingertips without rear-magnet-induced bending. The increased opening generated by the deflection of both fingers can be approximated as

$$d_{\text{open}} = d_0 + 2\delta_m + 2l_e \sin(\theta_m), \tag{8}$$

where l_e is the effective distance from the flexure region to the LCLE contact face. This expression shows that the available grasping region increases due to both lateral flexure deflection and angular fingertip rotation.

For a circular handle of diameter D_h, the basic geometric condition for insertion before closure can be written as

$$D_h \leq d_{\text{open}} - 2c_s, \tag{9}$$

Here, c_s represents the clearance allowance required for initial insertion. Equation (9) approximates direct insertion without additional finger deformation; compliant fingers may accommodate larger handles through further bending and changes in contact posture.

C. Handle-diameter eccentricity and opening-side transition

During manipulation, the handle diameter affects not only the initial grasping region, but also the way in which the LCLE–LCLE interface tends to open under load. As illustrated in Fig. 3, thin handles are positioned lower within the grasping region, whereas wider handles are positioned higher. As a result, the load transmitted by the object acts with a different eccentricity relative to the LCLE–LCLE interaction line.

Let y_n be the vertical position of the neutral LCLE–LCLE interaction line, and let $y_c(D_h)$ be the vertical position of the handle center for a handle diameter D_h. The load eccentricity is defined as

$$e(D_h) = y_c(D_h) - y_n, \tag{10}$$

where $e(D_h) < 0$ indicates that the load acts below the neutral interaction line, and $e(D_h) > 0$ indicates that it acts above it. Therefore, the sign of $e(D_h)$ determines the side from which the LCLE–LCLE interface tends to open during pulling.

The corresponding load-induced moment is written as

$$M_o(D_h) = F_o e(D_h), \tag{11}$$

where F_o is the force transmitted by the grasped object during manipulation. When $e(D_h) < 0$, the moment tends to open the LCLE–LCLE interface from the lower side. In contrast, when $e(D_h) > 0$, the moment tends to open the interface from the upper side.

For a circular handle, the vertical position of the handle center can be approximated from the geometry of two symmetric contact points. If the effective half-span between the two contact points is denoted by a, the handle-center position can be written as

$$y_c(D_h) = y_p + \sqrt{\left(\frac{D_h}{2}\right)^2 - a^2}, \tag{12}$$

where y_p is the vertical position of the contact-point reference. This expression is valid when the handle radius is larger than the effective half-span, i.e., $D_h/2 \geq a$. In practice, a can vary slightly with finger posture and handle diameter, but this approximation provides a useful geometric interpretation of the transition between lower-side and upper-side opening.

Substituting (12) into (10) gives a diameter-dependent eccentricity. A critical diameter D_{cr} can be defined by the condition $e(D_{cr}) = 0$. If a is locally approximated as constant around the transition region, the critical diameter is

$$D_{cr} = 2\sqrt{a^2 + (y_n - y_p)^2}. \tag{13}$$

The critical diameter D_{cr} represents the theoretical transition between lower-side and upper-side opening. Its value depends on the effective contact span a and the relative positions y_n and y_p, which may vary with finger deformation and grasping posture. Thus, the model provides a qualitative interpretation, while numerical estimation of D_{cr} requires characterization of the actual contact geometry.

The eccentricity effect can be incorporated into the object-load deflection term by writing

$$\delta_o(D_h) = \delta_{o,v} + \frac{M_o(D_h) L_f^2}{2E_f I_f}, \tag{14}$$

or equivalently,

$$\delta_o(D_h) = \frac{(F_o/2) L_f^3}{3E_f I_f} + \frac{F_o e(D_h) L_f^2}{2E_f I_f}. \tag{15}$$

The first term represents the deflection caused by the vertical object load, while the second term represents the additional bending contribution caused by the diameter-dependent load eccentricity. Since $e(D_h)$ is signed, this second term also captures the change in opening tendency between thin and wide handles.

Finally, substituting (15) into (2) gives the total finger deflection during manipulation:

$$\delta_f(D_h) = \delta_m + \frac{(F_o/2) L_f^3}{3E_f I_f} + \frac{F_o e(D_h) L_f^2}{2E_f I_f}. \tag{16}$$

This expression separates the pre-grasp expansion generated by the rear-pole magnet from the load-induced bending that occurs during manipulation. It also shows that handle diameter affects the final contact posture not only through insertion geometry, but also through the eccentricity of the transmitted load.

IV. LCLE–LCLE INTERACTION AND HANDLE GEOMETRY

A. Effective interaction area and gap

The electromagnetic holding force depends strongly on the interaction condition between the opposing LCLE cores. Although the handle itself does not need to be ferromagnetic, its geometry can modify the relative posture of the LCLE fingertips, changing the effective LCLE–LCLE interaction area and the effective gap.

The nominal electromagnetic force can be expressed in simplified form as

$$F_{\mathrm{em}} \propto \frac{B^2 A_{\mathrm{eff}}}{2\mu_0}, \tag{17}$$

where B is the magnetic flux density, A_{eff} is the effective LCLE–LCLE interaction area, and μ_0 is the permeability of free space. Since A_{eff} depends on object geometry, the handle size and shape directly influence the available magnetic holding force.

A more geometry-oriented expression can be written as

$$F_{\mathrm{em}}(D_h) \approx K \frac{A_{\mathrm{eff}}(D_h, \theta_c)}{g_{\mathrm{eff}}(D_h, \theta_c)^2}, \tag{18}$$

where D_h is the handle diameter, θ_c is the contact angle, g_{eff} is the effective air gap, and K groups the electromagnetic parameters associated with the LCLE. This expression highlights that a larger grasping region does not automatically imply larger holding force. A larger or differently shaped handle may increase the finger opening but reduce the effective LCLE–LCLE interaction area or increase the effective air gap.

V. EXPERIMENTAL EVALUATION OF HANDLE GEOMETRY

This section evaluates how handle geometry affects measured weight and pulling force during grasping. The experiments verify that handle diameter and shape modify contact posture, load-transfer path, and sensor reading, which is important for electromagnetic grippers since weight estimation is used to adjust the input energy required for stable grasping. Each handle-size configuration was tested four times before proceeding to the next size. The repetitions were used to verify

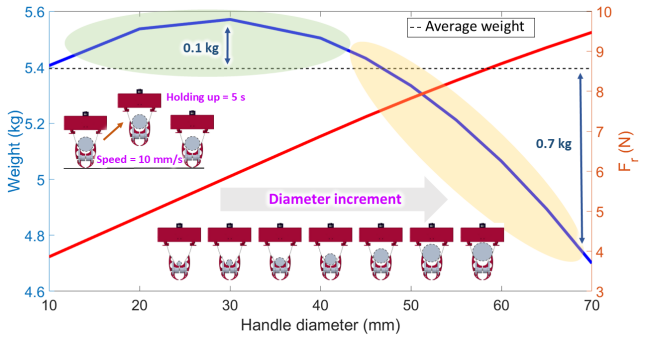


Fig. 4. Effect of circular-handle diameter on measured weight and pulling force. The variables illustrated in the figure indicate the experimental contact geometry, while the analytical formulation represents this effect through the handle diameter D_h, the load eccentricity $e(D_h)$, and the load-induced moment $M_o(D_h)$. The measured weight remains close to the average value for small and intermediate diameters, while the pulling force increases with handle diameter.

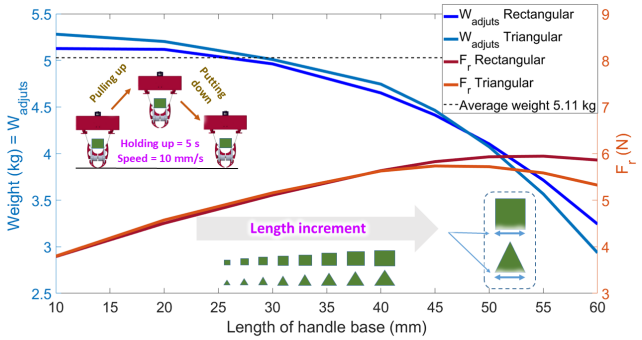


Fig. 5. Measured weight and pulling force for rectangular and triangular handles with different base lengths. The geometric labels in the figure illustrate the experimental change in handle size, while the analysis generalizes this effect as a change in contact posture and load-transfer path.

successful grasping, while statistical variability in the force and weight measurements was not quantified.

A. Circular-handle diameter effect

To evaluate circular-handle diameter effects, experiments were performed using the same object while varying only the handle diameter. The object was lifted at 10 mm/s and held for 5 s, with an average weight of approximately 5.4 kg. As shown in Fig. 4, the measured weight remains close to the average value for diameters between 10 mm and approximately 52 mm, with a maximum positive deviation of about 0.1 kg. This indicates that the force-sensing mechanism can estimate the object weight with relatively small error in this diameter range, while the pulling force increases approximately linearly with handle diameter.

Despite the constant object weight, the pulling force increases from approximately 3.8 N to 9.4 N as the circular handle diameter increases from 10 mm to 70 mm. Larger handles deform the carbon-fiber flexures and shift the contact region toward the gripper palm, modifying the load-transfer lever arm and increasing the tendency of the fingers to open. This provides a mechanical explanation for the higher measured pulling force. Meanwhile, the measured weight decreases for diameters above approximately 52 mm, indicating that the altered contact configuration also affects sensor accuracy.

From a sensing perspective, the handle diameter directly affects weight-measurement accuracy. For circular handles, the gripper maintains relatively low measurement error over a broad range of diameters; handles between 10 mm and 52 mm remain below approximately 1.85% error. Therefore, circular handles provide the most suitable geometry for the subsequent estimation-based control analysis.

B. Rectangular and triangular handles

Additional experiments were conducted with rectangular and triangular handles of different base lengths to evaluate noncircular geometries. The lifting speed was kept at 10 mm/s, the object was held for 5 s, and the average object weight was approximately 5.11 kg. Fig. 5 shows the measured weight and pulling force as functions of handle base length, with the dashed line indicating the average object weight.

For rectangular and triangular handles, the measured weight decreases as the base length increases, with the lowest reading error occurring between approximately 20 mm and 35 mm. In this range, rectangular handles provide more stable load transfer, reaching a minimum error of about 0.18%, while triangular handles show a higher minimum error of approximately 3.21%.

As the handle base length approaches 60 mm, weight-measurement errors increase to approximately 36.59% and 42.66% for rectangular and triangular handles, respectively. Sharp edges and corners modify load transfer and may cause mechanical interference or finger hooking, reducing grasp reliability. Since electromagnetic power regulation depends on the estimated object weight, such measurement errors may lead to inappropriate power commands. These results reveal the need for geometry-aware weight estimation and improved finger designs for noncircular handles.

These results confirm that noncircular handle geometries introduce sharper geometric transitions than circular ones. Rectangular and triangular handles do not produce a smooth contact migration; instead, they create discrete changes in contact location and load-transfer path. As a result, the measured weight becomes more sensitive to handle geometry, especially for larger handle sizes.

C. Selection of reference handle geometry

The previous results show that the accuracy of the measured weight is strongly affected by handle geometry. This is relevant since the estimated weight is used to determine the amount of electromagnetic input required to hold the object. If the weight reading is strongly distorted by the handle shape, the energy assignment may become inaccurate.

Among the tested geometries, circular handles provide the best compromise between grasping range, force variation, and sensing accuracy. Their gradual contact shift produces a

smoother load-transfer path, while rectangular and triangular handles introduce abrupt posture changes, larger reading errors, and greater sensitivity to size. Since cylindrical handles between 10 mm and 52 mm maintain a reading error below approximately 1.85%, they are selected as the reference geometry for the subsequent analysis. These results confirm that handle geometry affects contact posture, load eccentricity, required grasping force, and sensor-based weight estimation accuracy.

VI. Discussion

The proposed analysis shows that the rear-pole permanent magnets have a dual function: they attract the energized LCLEs to increase the initial fingertip separation and, through their connection to the carbon-fiber flexures, induce bending in the finger structure. This magnet-induced bending enlarges the available grasping region, allowing the gripper to accommodate larger handles without requiring the handle to be ferromagnetic.

The main contribution of the rear-pole magnets is the expansion of the effective grasping region, increasing the maximum magnetic capture distance from 23 mm to 50 mm (117%). This larger opening allows the gripper to accommodate handle sizes and geometries that would otherwise be inaccessible without the rear magnets, reducing insertion interference and facilitating successful grasp establishment. While this improvement expands the range of graspable objects, it does not necessarily increase the maximum electromagnetic holding force.

The 50 mm magnetic capture distance does not limit the maximum graspable handle diameter, as the flexible carbon-fiber fingers can accommodate circular handles up to 70 mm. However, larger handles alter fingertip alignment, load transfer, magnetic gap, and LCLE–LCLE interaction area, potentially reducing magnetic holding force despite the expanded grasping region.

The experimental results support this interpretation. Circular handles produce a gradual change in contact posture and maintain acceptable weight-measurement accuracy over a broad diameter range. Wider handles, however, reduce measurement accuracy and increase the pulling force required to maintain the grasp, which agrees with the load-eccentricity model. For non-circular handles, edges, corners, and abrupt contact transitions create less predictable load-transfer paths, making the measured weight more sensitive to handle size and shape. Therefore, handle geometry should be considered when evaluating electromagnetic grippers, especially for sensor-based weight estimation or power regulation.

VII. Conclusion

This paper presented a geometric and magnetic analysis of grasping-region expansion in electromagnetic grippers using rear-pole permanent magnets. The results show that the rear magnets push the LCLE fingertips forward and bend the carbon-fiber flexures, increasing the initial fingertip separation and enlarging the region available for mechanically grasping handles of different sizes and materials. In particular, the rear-pole magnet mechanism increased the maximum magnetic capture distance from approximately 23 mm to approximately 50 mm, corresponding to an expansion of about 117% before final object contact. This larger opening reduces insertion interference and decreases the likelihood of undesired fingertip deformation when positioning the gripper around the handle.

A simplified model related rear-magnet attraction, carbon-fiber deflection, finger opening, handle diameter, load eccentricity, and effective LCLE–LCLE interaction. The analysis shows that a larger grasping region can accommodate larger handles, but it can also change fingertip posture, load transfer, and magnetic interaction. Experiments with circular, rectangular, and triangular handles confirmed that handle geometry affects both the required grasping force and the accuracy of sensor-based weight measurement.

Future work will focus on improving finger geometry to reduce mechanical interference and developing geometry-aware weight estimation methods for reliable electromagnetic power regulation. Further experiments will also investigate flexure deformation and magnetic interaction under different grasping configurations.

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